Shikang Wen

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RESEARCH INTERESTS

- Carbon-neutral buildings
- AI-driven building design
- Building energy simulation
- Low-carbon HVAC systems

EDUCATION

Bachelor of Architecture (5-year professional degree), Shanghai Jiao Tong University — Sep 2018 – Jun 2023
PhD candidate, The Hong Kong Polytechnic University (Supervisor: Prof. Qingyan Chen) — Sep 2023 – Present (expected 2027)
Visiting Scholar, University of California, Berkeley — May 2026 – Present

ACADEMIC PUBLICATIONS

Journal papers (first author)

1. **S. Wen**, X. Hu, G. Hua, P. Xue, and D. Lai, Comparing the performance of four shading strategies based on a multi-objective genetic algorithm: A case study in a university library, 2022. (**Journal of Building Engineering, IF: 8.1**)
2. **S. Wen**, J. Xue, G. Hua, and D. Lai, Reconstructing fisheye luminance maps with a two-step network from a single low dynamic range image, 2024. (**Automation in Construction**, **IF: 12.6**)
3. **S. Wen**, R. You, and Q. Chen, Implementing building retrofitting strategies to halve campus office building carbon emissions by 2035: A case study in Hong Kong with techno-economic analysis, 2025. (**Energy and Buildings**, **IF: 8.0**)
4. **S. Wen**, B. Shen, Y. Yao, and Q. Chen, Energy performance and life-cycle analyses of advanced terminal devices and heat pumps for a medium-sized office building in different climate zones in China, 2025. (**Energy and Buildings**, **IF: 8.0**)
5. **S. Wen**, R. You, and Q. Chen, Energy-exergy performance and indoor comfort of a radiant ceiling cooling system with air-to-air sensible heat exchanger in hot and humid climate, 2025. (**Building and Environment**, **IF: 8.4**)
6. **S. Wen** and Q. Chen, An AI-assisted workflow for generative building design and energy modeling: Converting natural language design intent and images into 3D building energy model inputs, 2026 (online), Vol. 77, 2027. (**Advanced Engineering Informatics, IF: 11.5**)
7. **S. Wen**, R. You, and Q. Chen, An AI-driven design for energy-efficient buildings: A case study in a campus office building, 2026. (**Applied Energy, Under Review**)

Journal papers (co-author)

1. C. Zhang, F. Wang, Z. Wu, **S. Wen**, S.J. Cao, and Q. Chen, Techno-economic assessment of energy retrofitting in office buildings to achieve building carbon neutrality in Hong Kong, 2026. (**Preprint, SSRN 6848020**)

Conference papers

1. **S. Wen**, X. Hu, G. Hua, P. Xue, and D. Lai, A genetic-algorithm-based framework for assessing the performance of shading strategies, 2023. (**Building Simulation 2023: 18th Conference of IBPSA**)
2. **S. Wen** and Q. Chen, Field investigation of the impact of different terminal devices on thermal comfort and ventilation uniformity, 2026. (**IAQVEC 2026, E3S Web of Conferences**)

RESEARCH EXPERIENCE

Contribution in all projects: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft.

1. Optimization of daylit building spaces (Supervisor: Prof. Lai) — Mar 2021 – Jun 2022

Developed a multi-objective genetic-algorithm framework to compare four shading strategies for a university library; near-optimal designs reduced glare by up to 22% while maintaining daylight satisfaction.

2. Real-time glare detection (Supervisors: Prof. Lai and Prof. Xue) — Jul 2022 – Mar 2023

Developed a two-step deep-learning network that reconstructs fisheye luminance maps from a single low-dynamic-range image, running up to 95 times faster than the conventional HDR workflow.

3. Building retrofits in a high-density campus (Supervisors: Prof. Chen and Prof. You) Sep 2023 – May 2024

Evaluated retrofit pathways for Hong Kong campus office buildings using EnergyPlus and techno-economic analysis; combined strategies cut operational carbon emissions by about 50%.

4. Advanced radiant ceiling cooling system (Supervisor: Prof. Chen) May 2024 – Jan 2025

Designed and evaluated a radiant ceiling cooling system with an air-to-air sensible heat exchanger, reducing the indoor discomfort ratio from 54.0% to below 0.8% and energy use by 19.8% compared with a fan-coil system.

5. AI-driven energy-efficient building design (Supervisors: Prof. Chen and Prof. You) Feb 2025 – Jun 2025

Developed an AI-driven workflow that generates building designs and automatically converts them into EnergyPlus models.

6. AI-based 2D image to 3D EnergyPlus model (Supervisor: Prof. Chen) May 2025 – Nov 2025

Developed an image-to-3D workflow that converts a single building image into EnergyPlus geometry inputs, producing simulation-ready models in under five minutes.

7. AI-based scalable personal comfort HVAC control system (Supervisor: Prof. Chen) Dec 2025 – Sep 2026

Used CFD simulations to develop personal comfort HVAC control strategies that scale to building-level applications.

WORK EXPERIENCE

1. Intern, Shanghai Liben Architectural Design Office Dec 2020 – Feb 2021

Supported the lead designer in shaping the design of a school and independently handled the interior design of its cafeteria and dormitory.

2. Intern, East China Architectural Design and Research Institute Sep 2021 – Dec 2021

Participated in the design of the Urumqi Airport Terminal and the Hohhot Airport Terminal; under the guidance of the lead architect, designed the corridor bridge connecting the terminal building to the parking area.

TEACHING EXPERIENCE

1. Teaching Assistant, Sustainable Buildings, The Hong Kong Polytechnic University 2023/24 academic year
2. Teaching Assistant, HVACR, The Hong Kong Polytechnic University 2025/26 academic year

HONORS & AWARDS

SINOPEC “Cleaner Energy Better Life” Research Student Award (Merit) Aug 2026

PolyU International Collaborative Research Fellowship Jul 2026

IAQVEC 2026 Student Fellowship May 2026

National Inspirational Scholarship; Shanghai Jiao Tong University Scholarship 2021, 2022

First Prize and Second Prize, Li Zhengdao Science and Art Competition 2019, 2020

Structure competitions: Third Prize (SJTU, twice), Second Prize (Tongji University) 2019 – 2020

PROFESSIONAL SERVICE

- **External reviewer** in Journal of Building Engineering (since 2022)
- **External reviewer** in Building and Environment (since 2025)

TALKS & PRESENTATIONS

- Invited to present on AI-driven building design at the WeLL Seminar, Center for the Built Environment (CBE), UC Berkeley. Oct 2026
- Invited to give a talk at Arizona State University (ASU) on AI in building energy modeling. Sep 2026
- Presented on radiant cooling systems in hot and humid climates at the BEEE Research Salon, PolyU. Feb 2026

SKILLS & LANGUAGES

- Tools: Python, Rhino/Grasshopper, Revit/BIM, EnergyPlus, CFD
- Languages: English (professional working proficiency), Chinese (native)